Traffic Rule Summary

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1 Introduction

This document serves as an overview about formalized traffic rules based on the German Road Traffic Regulation (StVO), the Vienna Convention on Road Traffic [1] (VCoRT), judicial decisions, and comments by lawyers.

The formalization process of traffic rules in temporal logic can be separated into four steps:

1. **Extracting rules from legal sources**: Based on a traffic rule, all related rules can be extracted from legal sources. Legal sources can contain judicial decisions, other traffic law sections, and consultancy of layers.
2. **Concretizing extracted rules**: The legal texts and judicial decisions are combined and concretized using natural language. As part of the concretization, the situation when the rule is applicable should be expressed. It can also be the case that a single traffic rule addresses different aspects and therefore each has to be considered separately, e.g., StVO $\S4(1)$ addresses the distance between vehicles and also restricts the braking of a vehicle. The use of natural language allows lawyers to evaluate the concretization so that consistency between the formalized rules for autonomous vehicles and the traffic rules for human traffic participants can be ensured.
3. **Extracting functions, predicates, and propositions**: We use predicates, functions, and atomic propositions for the evaluation of traffic situations, e.g., the allowed speed limit or the velocity of a preceding vehicle. These elements should be defined in a modular way such that they can be easily reused.
4. **Creating temporal logic formulas**: The extracted predicates, functions, and propositions have to be combined to a temporal logic formula matching the concretized sentence from 2).

The remainder of this paper is structured as follows. Sec. 2 introduces mathematical foundations. Sec. 3 presents the formalization of traffic rules. In Sec. 4 predicates and auxiliary functions for the temporal logic formulas are defined.

2 Preliminaries

2.1 Legal information

Please note that no official translated version of the StVO exist. The translation used for this paper is based on the *German Law Archive*[1]. The formalized rules in this work consider the ego vehicle as the autonomous vehicle from whose perception the rules are written. All other traffic participants can either be controlled by humans or can also be autonomous vehicles.

We assume that the considered interstates have no intersections, driving directions are structurally separated so that only lanes with the same driving direction are adjacent, and have right-hand traffic. However, our formalization can also be used for left-hand traffic. Please note that it is possible on German interstates that no speed limit exist. In the following sections, we differentiate main carriage ways, access ramps, exit ramps, and shoulder lanes as potential lane types and solid, dashed, broad solid, and broad dashed lines as possible lane markings.

2.2 Road Network

We describe our road network by lanelets, which are atomic, drivable road segments, geometrically represented by a left and right bound, where the bounds are represented by polylines [2]. The set of all lanelets in a road network is denoted by $\mathbb{L} \cup \{\perp\}$, where $\perp$ is the bottom element for the case that no lanelet exist. Subsequently, we refer to a single lanelet as l, where $l \in \mathbb{L}$. A lanelet can be described by attributes, where the set of all attributes is denoted by $\mathbb{A}$. For interstates, we have $\mathbb{A} = \{access\_ramp, exit\_ramp, main\_carriage\_way, shoulder\}$. The function $\mathrm{attr}(l)$ returns the attribute which is valid for a lanelet (cf. Fig. 1). Similar, the left and right bounds of a lanelet can have lanelet markings. We denote the set of possible lanelet markings as $\mathbb{LM} = \{solid, dashed, broad\_solid, broad\_dashed\}$. The functions $\mathrm{mark_l}(l)$ and $\mathrm{mark_r}(l)$ return the left and right lanelet marking of a lanelet. The functions $\mathrm{l_{lb}}(l)$, $\mathrm{l_{rb}}(l)$, $\mathrm{l_{ini}}(l)$, and $\mathrm{l_{fin}}(l)$ return the left bound, right bound, left and right initial points, and left and right final points of a lanelet. The functions

$$\mathrm{adj_l}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{lb}}(l) = \mathrm{l_{rb}}(l') \\ \perp & \text{otherwise} \end{cases} \tag{1}$$

$$\mathrm{adj_r}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{rb}}(l) = \mathrm{l_{lb}}(l') \\ \perp & \text{otherwise} \end{cases} \tag{2}$$

$$\mathrm{succ}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{fin}}(l) = \mathrm{l_{ini}}(l') \\ & \wedge\ \mathrm{attr}(l) = \mathrm{attr}(l') \\ \perp & \text{otherwise} \end{cases} \tag{3}$$

$$\mathrm{pre}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{ini}}(l) = \mathrm{l_{fin}}(l') \\ & \wedge\ \mathrm{attr}(l) = \mathrm{attr}(l') \\ \perp & \text{otherwise} \end{cases} \tag{4}$$

return the adjacent left, adjacent right, successor, and predecessor lanelet of l, respectively (cf. Fig. 1). The functions $\mathrm{speed\_limit}(ls)$ and $\mathrm{required\_speed}(ls)$, where $ls \subseteq \mathbb{L}$, return a set of maximum and minimum speed limits which a vehicle is allowed to drive on a provided set of lanelets. The functions $\mathrm{ori_l}(l, s_\mathrm{l})$ and $\mathrm{width_l}(l, s_\mathrm{l})$, where s_l is a longitudinal position, return the orientation of the reference path θ_Γ and the width of a lanelet at s_l, respectively.

[1] https://germanlawarchive.iuscomp.org/?p=1290

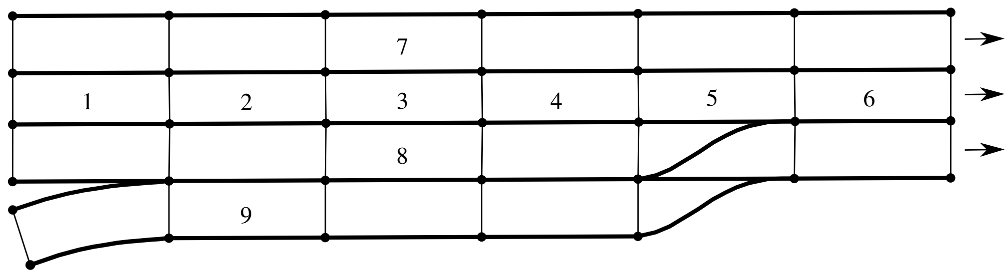

Figure 1: Road network defined by lanelets. The lanelets 1-6 form a lane. Lanelets 2, 4, 7, and 8 are the predecessor, successor, left adjacent, and right adjacent lanelet of lanelet 3, respectively. Lanelets 1-8 are of type $main\_carriage\_way$ and lanelet 9 is of type $access\_ramp$.

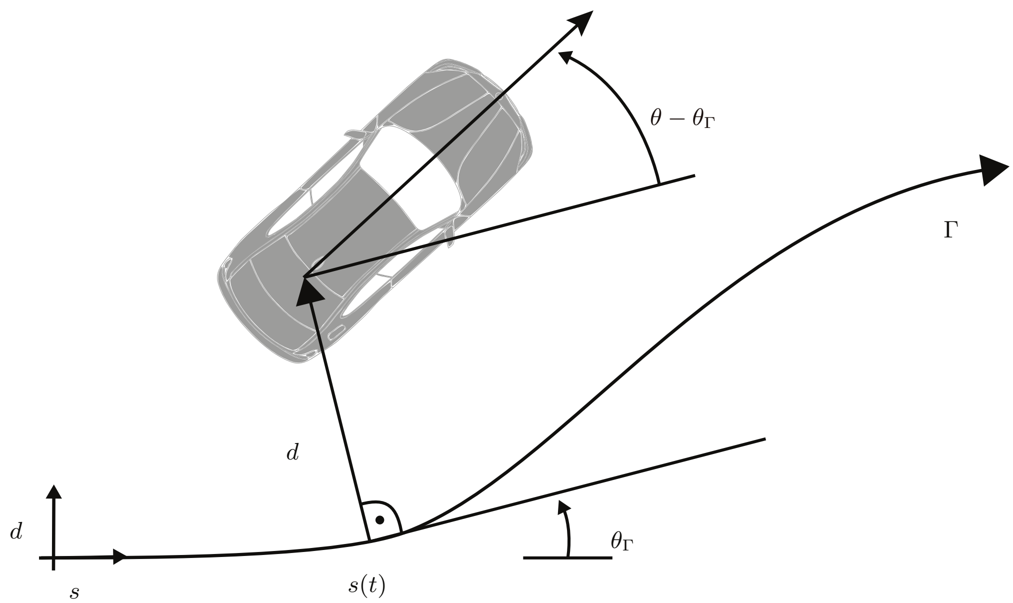

Figure 2: A curvilinear coordinate system aligned to the reference path Γ with orientation θ_Γ. The vehicle is described by the longitudinal position s, the lateral deviation of the reference path d, and the orientation θ.

Definition 1 (Lane)
The lane originating from a lanelet is defined recursively:

$$\begin{aligned}\mathrm{lane}(l) =& \big(\{\mathrm{succ}(l), l, \mathrm{pre}(l)\}\\ &\cup \mathrm{lane}(\mathrm{pre}(l)) \cup \mathrm{lane}(\mathrm{succ}(l))\big) \setminus \{\bot\}\end{aligned}$$

We use a curvilinear curvilinear coordinate system [3] as shown in Fig. 2.

2.3 Vehicle configuration

The state x of a vehicle consists of a longitudinal state $x_{\text{lon}} \in \mathbb{R}^n$ and a lateral state $x_{\text{lat}} \in \mathbb{R}^n$. The longitudinal state $x_{\text{lon}} = \begin{bmatrix} s & v & a & j \end{bmatrix}^T$ consists of position s, velocity v, acceleration a, and jerk j. The lateral state $x_{\text{lat}} = \begin{bmatrix} d & \theta \end{bmatrix}^T$ consists of the lateral distance to the reference path Γ and orientation θ (cf. Fig. 2). The input of the vehicle is denoted by $u(t)$. The underlying vehicle model can be arbitrary as long as a transformation to our state representation is possible. The occupancy of a vehicle for a given state is denoted by $\mathcal{O}(x(t))$. The functions $\mathrm{front}(x(t))$, $\mathrm{rear}(x(t))$, $\mathrm{right}(x(t))$, and $\mathrm{left}(x(t))$ return the position of the front bumper, rear bumper, rightmost point and leftmost point of a vehicle and $v_{\text{type}}(x(t))$ returns the maximum allowed velocity for a vehicle type, e.g., a truck. All variables associated to the ego vehicle are denoted by the subscript $\square_{\text{ego}}$, all variables associated to other vehicles are denote by the subscript $\square_{\text{o}}$, and arbitrary vehicles including the ego vehicle are denoted by the subscript $\square_{\mathrm{k}}$ and $\square_{\mathrm{p}}$, where $\mathrm{o} \in \{1...N\}$, N is the number of other traffic participants, and $\mathrm{k}, \mathrm{p} \in \{0...(N+1)\}$ represent all vehicles in the road network including the ego vehicle. For a given initial state x_0, an input trajectory $u(\cdot)$, we introduce the solution of the model $\dot{x} = f(x, u)$ of the ego vehicle over time t as $\xi(t; x_0, u(\cdot))$. We denote a braking input by $u_{\mathrm{brake}}(\cdot)$. The time at which a safe state is reached for $u(\cdot)$, given the system dynamics and x_0, is denoted by $t_s(u(\cdot))$. We consider a state to be safe on interstates for the ego vehicle if the ego vehicle is in standstill.

2.4 Finite Linear Temporal Logic

We use Finite Linear Temporal Logic (FLTL) since it is interpreted over finite traces of Boolean elements. Given is a set $\mathbb{AP}$ of atomic propositions, where each atomic proposition $\sigma_i \in \mathbb{AP}$ represents a Boolean statement. A FLTL formula ϕ is defined as

$$\begin{aligned} \phi ::= &\sigma_i \,|\, \neg\phi \,|\, \phi_1 \wedge \phi_2 \,|\, \phi_1 \vee \phi_2 \,| \\ &X(\phi) \,|\, \overline{X}(\phi) \,|\, \phi_1 U \phi_2 | \, G(\phi) | \, F(\phi) \end{aligned}$$

where the letter X stands for the weak next operator, $\overline{\mathrm{X}}$ for the strong next, U for the until, G for the globally, and F for the future operator. We also require the Boolean operators $\neg$, $\wedge$, and $\vee$. Unary connectives have precedence over binary connectives, $\neg$, X, and $\overline{X}$ have equal precedence and U has precedence over $\wedge$ and $\vee$. The implication $a \implies b$ is defined as $\neg a \vee b$.

We describe the semantics of FLTL informally. For a detailed description of the semantics we refer the reader to [4] and [5]. The weak next operator expresses that if a next state exists then this state has to satisfy ϕ, whereas the strong next operator expresses that the next state has to exist and that this state has to satisfy ϕ. The until operator says that ϕ_1 is true at least until ϕ_2 is true. The globally operator states that ϕ holds at all states and the future operator states that ϕ holds at some future state.

Table 1: Overview about the formalized traffic rules. For brevity we do not show the parameters of the predicates.

Rule name	Law reference	FLTL formula
R_G3	StVO §3 (1), §3 (3), §18 (1), §18 (5), §18 (6), traffic sign 274	$\mathrm{G}(\text{keeps\_lane\_speed\_limit} \wedge \text{keeps\_fov\_speed\_limit} \wedge \text{keeps\_braking\_spe}$ $\wedge\, \text{keeps\_road\_condition\_speed\_limit} \wedge \text{keeps\_type\_speed\_limit})$
R_G4	StVO traffic sign 275	$\mathrm{G}(\text{keeps\_min\_speed\_limit})$
R_I5	StVO §1 (2), §18 recital 11	$\mathrm{G}(\text{on\_access\_ramp} \wedge \mathrm{F}(\text{on\_main\_carriage\_way})$ $\implies \neg(\neg\, \text{in\_rightmost\_lane} \wedge \mathrm{F}(\text{in\_rightmost\_lane})))$

3 Traffic rule formalization

We formalize selected rules from the StVO, e.g., establishing an emergency lane or keeping a safe distance, which are necessary to drive on German interstates and also provide their textual concretization as well as additional comments. The temporal logic formula and the related laws and traffic signs for each rule are listed in Tab. 1[2]. Predicates and functions required for the FLTL formulas are introduced in Sec. 4. We first present general rules and afterward interstate-specific rules. The rules are written from the viewpoint of the autonomous ego vehicle.

- The autonomous mode/monitor is deactivated in case of emergencies

3.1 General traffic rules

3.1.1 R_G1 - Safe distance to preceding vehicle

Law reference:

- **StVO §4 (1)** A person operating a vehicle moving behind another vehicle must, as a rule, keep a sufficient distance from that other vehicle so as to be able to pull up safely even if it suddenly slows down or stops.
- **VCoRT §13 (5):** The driver of a vehicle moving behind another vehicle shall keep at a sufficient distance from that other vehicle to avoid collision if the vehicle in front should suddenly slow down or stop.

Concretization: The ego vehicle following vehicles within the same lane has to keep a safe distance which ensures collision freedom to each of these vehicles, even if one or several of them are suddenly stopping.

LTL formula: $\mathrm{G}(\text{is\_in\_same\_lane} \wedge \text{in\_front\_of} \implies \text{keeps\_safe\_distance\_prec})$

Comment: The rule has to be evaluated for every other traffic participant within the sensor range. Vehicles outside are taken care of by the allowed velocity in rule R_G3.

3.1.2 R_G2 - Unnecessary braking

Law reference:

- **StVO §4 (1):** The person operating the vehicle in front must not brake suddenly without a compelling reason.
- **VCoRT §17 (1):** No driver of a vehicle shall brake abruptly unless it is necessary to do so for safety reasons.

[2] The terms *OLG* and *Bay VRS* are abbreviations for German types of courts.

Concretization: The ego vehicle is not allowed to brake abruptly without reason. The ego vehicle brakes abruptly if: a) The ego vehicle brakes although there is no leading vehicle and there is no reason caused by a traffic regulation to reduce the velocity. b) There is an obstacle in front of the vehicle and the acceleration difference and jerk difference do not violate predefined thresholds. c) The maximum allowed speed will change or has changed due to unrecognized traffic signs or environmental conditions. All possible braking situations which are not covered by the three cases are unnecessary.

LTL formula: $\mathrm{G}(\neg\,\mathrm{unnecessary\_braking})$

3.1.3 R_G3 - Maximum speed limit

Law reference:

Concretization: The ego vehicle is not allowed to exceed the speed limit of the lanes it drives on, the speed limit which is necessary to ensure collision freedom with respect to vehicles outside the field of view, the speed which is necessary to comfortably react to traffic regulations, the speed limit based on road conditions, and the maximum velocity which is allowed for its vehicle type.

3.1.4 R_G4 - Minimum speed limit

Law reference:

Concretization: The ego vehicle has to consider the minimum speed limit traffic sign.

3.1.5 R_G5 - Traffic flow

Law reference: StVO §3 (2)

- **StVO §3 (2):** No motor vehicle must, without good reason, travel so slowly as to impede the flow of traffic.

Concretization: The ego vehicle is not allowed to travel so slowly that it impedes traffic flow.

LTL formula: $\mathrm{G}(\mathrm{preserves\_traffic\_flow})$

3.2 Interstate-specific traffic rules

3.2.1 R_I1 - Stopping

Law reference:

- **StVO §12 (1)** Stopping is not permitted: [...] on acceleration and deceleration lanes; [...]
- **StVO §18 (8)** Stopping is prohibited, including on verges.
- **OLG Karlsruhe DAR 02, 34; OLG Frankfurt DAR 01, 504:** Stopping is prohibited on all parts that serve the purpose of smooth traffic.
- **OLG Hamm VersR 91, 83:** Stopping is prohibited on the emergency lane.

- **Bay VRS 59, 54 = StVE 23:** Stopping is prohibited on access lanes from and to parking spots.
- **Burmann, et al., Straßenverkehrsrecht, § 18 StVO, recital 23**: If an accident occurs, the vehicles involved must stop pursuant to StVO Sec. §11 (2) and the persons involved have to wait in order to ascertain identification.
- **OLG Köln VM 74:** Stopping on the emergency lane is only allowed temporarily and in emergency cases. Broken down cars must be removed from the emergency lane as fast as possible.

Concretization: If the ego vehicle is not part of a congestion or the directly leading vehicle is not standing, then stopping on the main carriage way, shoulder lane, and ramps is forbidden.

Comment: The last two items are currently not considered. What is in case of a congestion? standing leading vehicle? We allow the ego vehicle to stop if a leading vehicle also stops. Vehicles may stop on the marked parking areas themselves, or at the parking areas at the restaurants or gas stations.

LTL formula: $\mathrm{G}(\neg(\mathrm{in\_congestion} \vee \mathrm{exist\_standing\_leading\_vehicle}) \implies \neg\,\mathrm{in\_standstill})$

3.2.2 R_I2 - Overtaking on the right-hand side

Law reference:

- **BGHSt 12, 258, 260:** Overtaking on the right-hand side is generally prohibited, except for the following cases.
- **StVO §7 (2):** If the density of traffic has resulted in queues of vehicles in the lanes of one carriageway, traffic on the right (nearside lane, middle lane) may move faster than traffic on the left (offside lane, middle lane).
- **StVO §7 (2a)h:** If, on a carriageway for one direction of traffic, a vehicle queue has formed and come to a standstill or is moving at low speed in the left-hand lane, vehicles may overtake this queue on the right at a slightly higher speed and with the utmost care.
- **StVO §7a (1):** Where lanes diverge from the main carriageway, especially on motorways and motor roads, vehicles turning off may, from the beginning of and to the right of a lane marking with broad lines (sign 340), travel faster than traffic on the main carriageway.
- **StVO §7a (2):** On motorways and other roads outside built-up areas, vehicles may travel faster in acceleration lanes than traffic on the main carriageway.
- **StVO §7a (3):** In deceleration lanes, vehicles must not travel faster than traffic on the main carriageway. If traffic on the main carriageway is moving slowly or is stationary, vehicles in a deceleration lane may overtake at a moderate speed and with great care.
- **StVO §OLG Frankfurt/M. VRS 63, 386; OLG Düsseldorf VRS 78, 473:** In the range of guiding signs, e.g. Sign 448, if vehicles turn right and keep their lane when turning.

Concretization: The ego vehicle is not allowed to drive faster than any vehicle on lanes left of it. An exception is, if a) the vehicle in a left lane is in a congestion; b) the vehicle in the left lane is in a congestion on the main carriage way and the ego vehicle is on an off ramp; c) the left vehicle and the ego vehicle are separated by a broad lane marking; d) the ego vehicle is within an access ramp; d) the ego vehicle is close to a guiding sign and follows the guiding sign. For a) and b), the ego vehicle is allowed to drive faster than the vehicle in the left lane with only a slightly higher speed.

Comment: The second case of R_I2 is covered in the first clause of the following FLTL formula. The fourth case of R_I2 allows to drive on access ramps with higher speed than on the main carriage way. However, if a congestion is on the left side of the ego vehicle and the ego vehicle is on an access ramp, it is only allowed to drive with slightly higher speed. This rule has to be evaluated with respect to every other vehicle within the sensor range. Low speed means in this context a maximum speed of 60 km/h (comp. Burmann, et al., Straßenverkehrsrecht, StVO §(5) , recital 58)and slightly higher speed means 20 km/h faster than vehicles on the left lanes (BGHSt 22, 137; Bay 77, 172 = VRS 54, 212; OLG Köln VRS 61, 45). We currently do not consider the exception for guiding signs.

LTL formula: $\mathrm{G}((\mathrm{congestion\_left} \wedge \mathrm{drives\_with\_slightly\_higher\_speed}) \vee (\mathrm{right\_of\_broad\_lane\_marking} \wedge \mathrm{left\_of\_broad\_lane\_marking}) \quad \vee (\mathrm{on\_access\_ramp} \wedge \neg \mathrm{congestion\_left}) \vee \neg \mathrm{drives\_faster\_than\_vehicle\_left})$

3.2.3 R_I3 - No reversing and turning

Law reference:

- **StVO §18 (7):** Making U-turns and reversing are prohibited.

Concretization: Making U-turns and reversing is prohibited.

Comment: U-turns and reversing is prohibited for all lane types (comp. Bay VM 80, 47; OLG Hamm VRS 59, 458). U-turning can be justified in exceptional circumstances, e.g. colliding with another vehicle at night (BGH VersR 60, 802; VRS 14, 89) or being the safest means of the averting of dangers. Reversing can exceptionally be justified as an emergency measure, e.g., while losing an item of luggage (comp. Burmann, et al., Straßenverkehrsrecht, StVO §(18), recital 21).

LTL formula: $\mathrm{G}(\neg \mathrm{makes\_u\_turn} \wedge \neg \mathrm{reverses})$

3.2.4 R_I4 - Emergency lane

Law reference:

- **StVO §11 (2) - English:** As soon as vehicles on motorways and roads outside a built-up area with at least two lanes for one direction start to move at walking pace or come to a standstill, these vehicles must leave a gap for one direction between the lane on the far left and the lane immediately adjacent to it on the right to allow police and emergency vehicles to pass.
- **StVO §11 (2) - German:** Sobald Fahrzeuge auf Autobahnen sowie auf Außerortsstraßen mit mindestens zwei Fahrstreifen für eine Richtung mit Schrittgeschwindigkeit fahren oder sich die Fahrzeuge im Stillstand befinden, müssen diese Fahrzeuge für die Durchfahrt von Polizei- und Hilfsfahrzeugen zwischen dem äußerst linken und dem unmittelbar rechts daneben liegenden Fahrstreifen für eine Richtung eine freie Gasse bilden.

Concretization: A vehicle in a congestion has to drive leftmost if it is in the leftmost lane and has to drive rightmost otherwise. If the interstate is not broad enough to create an emergency lane, all vehicles should drive in the rightmost lane.

LTL formula: $\mathrm{G}(\mathrm{in\_congestion} \implies ((\mathrm{interstate\_broad\_enough} \implies ((\mathrm{in\_leftmost\_lane} \implies \mathrm{drives\_leftmost}) \wedge (\neg \mathrm{in\_leftmost\_lane} \implies \mathrm{drives\_rightmost}))) \wedge (\neg \mathrm{interstate\_broad\_enough} \implies \mathrm{drives\_rightmost})))$

3.2.5 R_I5 - Consider entering vehicles

Law reference:

Concretization: The ego vehicle is not allowed to move from the main carriage way to the rightmost lane if another vehicle is about to enter the main carriage way from the access ramp.

LTL formula:

4 Predicates and Functions

The predicates and functions necessary to describe the presented traffic rules are categorized in position, velocity, braking, and general elements. We use first-order logic to define the predicates and functions (where the atomic propositions of the FLTL formulas correspond to the defined predicates).

4.1 Position Elements

The occupied lanelets of a state x_k are defined by the intersection of its occupancy with the corresponding lanelets:

$$\mathrm{occupied\_lanelets}(x_\mathrm{k}) = \big\{l \in \mathbb{L} \,\big|\, l \cap \mathcal{O}(x_\mathrm{k}) \neq \emptyset\big\}.$$

The lanes which a vehicle occupies are defined as:

$$\mathrm{lanes}(x_\mathrm{k}) = \{\mathrm{lane}(l) \,|\, l \in (\mathrm{occupied\_lanelets}(x_\mathrm{k}))\}.$$

This information is used to reason whether the k^{th} vehicle is in any lane the p^{th} vehicle is in:

$$\mathrm{is\_in\_same\_lane}(x_\mathrm{k}, x_\mathrm{p}) \iff \mathrm{lanes}(x_\mathrm{k}) \cap \mathrm{lanes}(x_\mathrm{p}) \neq \emptyset.$$

The k^{th} vehicle is in front of the p^{th} vehicle if its longitudinal position is larger than the one of the p^{th} vehicle:

$$\mathrm{in\_front\_of}(x_\mathrm{p}, x_\mathrm{k}) \iff \mathrm{front}(x_\mathrm{p}) \leq \mathrm{rear}(x_\mathrm{k}).$$

The information whether a vehicle is on a the main carriage way, is given by

$$\begin{aligned}
&\mathrm{on_main_carriage_way}(x_\mathrm{k}) \iff \\
&\quad main_carriage_way \in \\
&\qquad \{\mathrm{attr}(l) \,|\, l \in \mathrm{occupied_lanelets}(x_\mathrm{k})\}.
\end{aligned}$$

Other lanelet attributes can be evaluated similarly. The k^{th} vehicle is left of the p^{th} vehicle if

$$\begin{aligned}
&\mathrm{vehicle_is_left}(x_\mathrm{k}, x_\mathrm{p}) \iff \\
&\quad \big(\mathrm{rear}(x_\mathrm{p}) \leq \mathrm{rear}(x_\mathrm{k}) \leq \mathrm{front}(x_\mathrm{p}) \\
&\quad \vee (\mathrm{front}(x_\mathrm{p}) < \mathrm{front}(x_\mathrm{k}) \wedge \mathrm{rear}(x_\mathrm{k}) < \mathrm{rear}(x_\mathrm{p})) \\
&\quad \vee \mathrm{rear}(x_\mathrm{p}) \leq \mathrm{front}(x_\mathrm{k}) \leq \mathrm{front}(x_\mathrm{p})\big) \wedge d_\mathrm{k} > d_\mathrm{p},
\end{aligned}$$

evaluates to true. To determine whether a vehicle is right of a broad lane marking, the following predicate is introduced:

$$\begin{aligned}
&\mathrm{right_of_broad_lane_marking}(x_\mathrm{k}) \iff \\
&\big\{l \in \mathrm{occupied_lanelets}(x_\mathrm{k}) \,\big|\, \mathrm{mark}_\mathrm{r}(l) \in \{broad_solid, \\
&\quad broad_dashed\}\big\} = \emptyset \wedge \\
&\big\{l \in \mathrm{lanelets_left_of_vehicle}(x_\mathrm{k}) \,\big| \\
&\quad \mathrm{mark}_\mathrm{r}(l) \in \{broad_solid, broad_dashed\}\big\} \neq \emptyset,
\end{aligned}$$

where

$$\begin{aligned}
&\mathrm{lanelets_left_of_vehicle}(x_\mathrm{k}) = \\
&\bigcup\{\mathrm{lanelets_left_of_lanelet}(l) \,|\, l \in \mathrm{occupied_lanelets}(x_\mathrm{k})\},
\end{aligned}$$

returns the set containing all lanelets left of the vehicle and the recursively defined function

$$\begin{aligned}&\text{lanelets\_left\_of\_lanelet}(l) = \\ &\quad \{\text{adj}_\text{l}(l)\} \cup \text{lanelets\_left\_of\_lanelet}(\text{adj}_\text{l}(l)),\end{aligned}$$

returns all lanelets left of a given lanelet, where $\text{lanelets\_left\_of\_lanelet}(\perp) = \emptyset$. Similarly, the predicates $\text{left\_of\_broad\_lane\_marking}$, $\text{lanelets\_right\_of\_vehicle}$, and $\text{lanelets\_right\_of\_lanelet}$ can be defined. The width of a road given a lanelet and longitudinal position s_k is recursively defined as:

$$\text{road\_width}(l, s_\text{k}) = \sum_{l' \in \text{adj\_lanelets}(l)} \text{width}_\text{l}(l', s_\text{k})$$

where $\text{road\_width}(\perp, s_\text{k}) = 0$ and

$$\begin{aligned}\text{adj\_lanelets}(l) = \big(&\{l\} \cup \text{adj\_lanelets}(\text{adj}_\text{r}(l)) \\ &\cup \text{adj\_lanelets}(\text{adj}_\text{l}(l))\big) \setminus \{\perp\}.\end{aligned}$$

If the width of a road is larger than a certain threshold w_lane^{min}, the road is declared as broad enough:

$$\begin{aligned}&\text{interstate\_broad\_enough}(x_\text{k}) \iff \\ &\quad \forall l \in \text{occupied\_lanelets}(x_\text{k}) : \text{road\_width}(l, s_\text{k}) > w_\text{lane}^{min}.\end{aligned}$$

A vehicle is in the leftmost lane if any of its occupied lanelets has no left adjacent lanelet:

$$\begin{aligned}&\text{in\_leftmost\_lane}(x_\text{k}) \iff \\ &\quad \exists l \in \text{occupied\_lanelets}(x_\text{k}) : \text{adj}_\text{l}(l) = \perp.\end{aligned}$$

The proposition in_rightmost_lane for the rightmost lane is defined analogously. To evaluate if a vehicle drives as leftmost or as rightmost as possible, the lateral distance from the lane center and the lane width are used:

$$\begin{aligned}&\text{drives\_leftmost}(x_\text{k}, s_\text{k}) \iff \forall l \in \text{occupied\_lanelets}(x_\text{k}) : \\ &\left(\frac{1}{2}\text{width}_\text{l}(l, s_\text{k}) + \text{left}(x_\text{k})\right) \le d_\text{bound},\end{aligned}$$

where drives_rightmost is analogously defined.

4.2 Velocity Elements

The speed limit regulated by traffic signs is the minimum speed limit of all lanelets a vehicle occupies:

$$v_\text{sl}^\text{max}(x_\text{k}) = \min\big(\text{speed\_limit}(\text{occupied\_lanelets}(x_\text{k}))\big).$$

Similar, the minimum speed limit is defined as:

$$v_\text{sl}^\text{min}(x_\text{k}) = \max\big(\text{required\_speed}(\text{occupied\_lanelets}(x_\text{k}))\big).$$

The ego vehicle has to be able to stop within the field of view. Therefore, the maximum allowed velocity based on the field of view v_fov is chosen such that the ego vehicle can come safely to a standstill even when a standing vehicle enters the field of view fov. For this, we search for the maximum value of v_fov which fulfills the following condition:

$$\text{front}(\xi(t_s(u_\text{brake}(\cdot)); x_\text{lon}^\text{sr}, u_\text{brake}(\cdot))) \le fov,$$

where $x_{\text{lon}}^{\text{sr}} = [0, v_{\text{fov}}, a_{\text{ego}}^{\max}, j_{\text{ego}}^{\max}]^T$ is the worst-case initial state. Additionally, the ego vehicle has to reduce its velocity to an upcoming speed limit such that it does not brake abruptly. Therefore, the speed limit v_{br} is introduced which ensures that a trajectory for the velocity reduction exists which does not exceed a predefined acceleration threshold $a_{\text{abrupt}} < 0$ and jerk threshold $j_{\text{abrupt}} < 0$. We calculate v_{br} analogously to v_{fov}, where the following two conditions have to be fulfilled:

$$\begin{aligned}
&x_{\text{lon}}^{\text{f}} = \xi(t^*; x_{\text{lon}}^0, u(\cdot)),\\
&\forall t \in [t_0, t^*] : a_{\text{ego}}(t) \geq a_{\text{abrupt}} \wedge j_{\text{ego}}(t) \geq j_{\text{abrupt}},
\end{aligned}$$

where $x_{\text{lon}}^0 = [0, v_{\text{br}}, a_{\text{ego}}^{\max}, j_{\text{ego}}^{\max}]^T$ and $x_{\text{lon}}^{\text{f}} = [fov, v_{\text{sl}}^*, 0, 0]^T$ are the worst-case initial state and the final state at which the upcoming speed limit v_{sl}^* is reached. Additionally, we assume that the ego vehicle has to consider the maximum allowed velocity v_{w} based on the road conditions for which we assume that it is externally provided. A vehicle keeps the speed limit given by traffic signs if its velocity is less than $v_{\text{sl}}^{\max}$:

$$\text{keeps\_lane\_speed\_limit}(x_{\text{k}}, v_{\text{k}}) \iff v_{\text{k}} \leq v_{\text{sl}}^{\max}(x_{\text{k}}).$$

The other speed limit predicates are defined analogously. A vehicle only needs to consider the minimum required speed limit if its velocity is below v_{fov}, v_{w}, and v_{type}:

$$\begin{aligned}
&\text{keeps_min_speed_limit}(x_{\text{k}}, v_{\text{k}}) \iff\\
&v_{\text{sl}}^{\min}(x_{\text{k}}) < \min(v_{\text{fov}}, v_{\text{w}}, v_{\text{type}}) \implies v_{\text{sl}}^{\min}(x_{\text{k}}) \leq v_{\text{k}}.
\end{aligned}$$

To ensure high traffic throughput on interstates, vehicles must not impede traffic flow:

$$\begin{aligned}
&\text{preserves_traffic_flow}(x_{\text{k}}, v_{\text{k}}) \iff\\
&\quad \neg\,\text{slow_leading_vehicle} \implies \text{v_max}_1(x_{\text{k}}) - v_{\text{k}} > \Delta v_{\text{fl}},
\end{aligned}$$

where $\text{v\_max}_1(x_{\text{k}}) = \min(v_{\text{br}}, v_{\text{fov}}, v_{\text{w}}, v_{\text{sl}}^*(x_{\text{k}}), v_{\text{type}}(x_{\text{k}}))$,

$$v_{\text{sl}}^*(x_{\text{k}}) = \begin{cases} v_{\text{sl}}^{\max}(x_{\text{k}}) & \text{if } v_{\text{sl}}^{\max}(x_{\text{k}}) \neq \inf,\\ v_{\text{su}} & \text{otherwise,} \end{cases}$$

v_{su} is the suggested velocity which is assumed if no speed limit is present,

$$\begin{aligned}
&\text{slow_leading_vehicle}(x_{\text{k}}) \iff\\
&\quad \exists \text{o} \in \{1...N\} : \text{v_max}_2(x_{\text{o}}) - v_{\text{o}} > \Delta v_{\text{fl}}\\
&\quad \wedge \text{is_in_same_lane}(x_{\text{k}}, x_{\text{o}}) \wedge \text{in_front_of}(x_{\text{k}}, x_{\text{o}}),
\end{aligned}$$

$\text{v\_max}_2(x_{\text{k}}) = \min(v_{\text{w}}, v_{\text{sl}}^*(x_{\text{k}}), v_{\text{type}}(x_{\text{k}}))$, and Δv_{fl} is a threshold which indicates if a vehicle drives slowly. A vehicle is in standstill if its velocity is close to zero:

$$\text{in\_standstill}(v_{\text{k}}) \iff -v_{\text{err}} \leq v_{\text{k}} \leq v_{\text{err}},$$

where v_{err} captures measurement uncertainties and is close to zero. This predicate can be used to evaluate if leading vehicles exist which are standing:

$$\begin{aligned}
&\text{exist_standing_leading_vehicle}(x_{\text{k}}) \iff\\
&\quad \exists \text{o} \in \{1...N\} : \text{is_in_same_lane}(x_{\text{k}}, x_{\text{o}})\\
&\quad \wedge \text{in_front_of}(x_{\text{k}}, x_{\text{o}}) \wedge \text{in_standstill}(x_{\text{o}}).
\end{aligned}$$

The k^{th} vehicle drives with slightly higher speed than the p^{th} vehicle if the following predicate is true:

$$\begin{aligned}
&\text{drives_with_slightly_higher_speed}(x_{\text{k}}, x_{\text{p}}) \iff\\
&\quad 0 < x_{\text{k}} - x_{\text{p}} < v_{\text{so}},
\end{aligned}$$

where v_{so} indicates slightly higher speed. The following proposition indicates that k^{th} vehicle drives faster than any vehicle on its left side:

$$\begin{aligned}&\text{drives\_faster\_than\_vehicle\_left}(v_{\text{k}}) \iff \\ &\quad \exists \text{o} \in \{1...N\} : \text{vehicle\_is\_left}(x_{\text{o}}) \wedge v_{\text{k}} > v_{\text{o}}.\end{aligned}$$

A backward driving vehicle has a negative velocity considering uncertainties:

$$\text{reverses}(v_{\text{k}}) \iff v_{\text{k}} < -v_{\text{err}}.$$

4.3 Braking Elements

The safe distance which has to be kept by the ego vehicle to a leading vehicle is based on the definitions in [6] and [7]:

$$d_{\text{safe}}(v_{\text{ego}}, v_{\text{o}}) = \frac{v_{\text{o}}^2}{2a_{\text{o}}^{\text{min}}} - \frac{v_{\text{r}}^2}{2a_{\text{ego}}^{\text{min}}} + v_{\text{ego}} t_{\text{d}} + \frac{1}{2} a_{\text{ego}}^{\text{max}} t_{\text{d}}^2,$$

where $v_{\text{r}} = v_{\text{ego}} + a_{\text{ego}}^{\text{max}} t_{\text{d}}$ and t_{d} denotes the reaction time of the ego vehicle. We assume that other vehicles can brake stronger than the ego vehicle: $a_{\text{o}}^{\text{min}} < a_{\text{ego}}^{\text{min}}$. The safe distance is kept if the relative distance between the vehicles is larger than the safe distance:

$$\begin{aligned}&\text{keeps\_safe\_distance\_prec}(x_{\text{ego}}, x_{\text{o}}, v_{\text{ego}}, v_{\text{o}}) \iff \\ &\quad d_{\text{safe}}(v_{\text{ego}}, v_{\text{o}}) < (\text{rear}(x_{\text{o}}) - \text{front}(x_{\text{ego}})).\end{aligned}$$

Unnecessary braking is defined based on the ego vehicle's acceleration and jerk:

$$\begin{aligned}&\text{unnecessary\_braking}(x_{\text{ego}}, a_{\text{ego}}, j_{\text{ego}}) \iff \\ &\quad (\neg\, \text{velocity\_reduction\_necessary} \wedge a_{\text{ego}} < 0) \implies \\ &\quad\; \big(\nexists \text{o} \in \{1...N\} : \text{in\_front\_of}(x_{\text{k}}, x_{\text{o}}) \\ &\quad\; \wedge a_{\text{ego}} < a_{\text{abrupt}} \wedge j_{\text{ego}} < j_{\text{abrupt}}\big) \\ &\quad\; \vee \big(\exists \text{o} \in \{1...N\} : \text{in\_front\_of}(x_{\text{k}}, x_{\text{o}}) \\ &\quad\; \wedge (a_{\text{ego}} - a_{\text{o}}) < a_{\text{abrupt}} \wedge j_{\text{ego}} < j_{\text{abrupt}}\big),\end{aligned}$$

where $\text{velocity\_reduction\_necessary}$ indicates if a braking maneuver is necessary due to an externally reason, e.g., another vehicle violates traffic rules so that the ego vehicle has to react to prevent a collision. We assume this information is provided by another system.

4.4 General Elements

The k^{th} vehicle is part of a congestion if there is a predefined number of slow driving vehicles in front of it:

$$\begin{aligned}&\text{in\_congestion}(x_{\text{k}}) \iff \\ &\quad \exists^{n_{\text{con}}}\, \text{o} \in \{1...N\} : \text{in\_front\_of}(x_{\text{k}}, x_{\text{o}}) \\ &\quad \wedge \text{is\_in\_same\_lane}(x_{\text{o}}) \wedge v_{\text{o}} \leq v_{\text{con}},\end{aligned}$$

where v_{con} indicates slow moving vehicles in a congestion and n_{con} is the minimum required number of vehicles which are part of a congestion within one lane. A congestion is left of a vehicle if some vehicle on the left side is in a congestion:

$$\begin{aligned}&\text{congestion\_left}(x_{\text{k}}) \iff \\ &\quad \exists \text{o} \in \{1...N\} : \text{vehicle\_is\_left}(x_{\text{k}}, x_{\text{o}}) \\ &\quad \wedge \text{in\_congestion}(x_{\text{k}}).\end{aligned}$$

For a turning vehicle the difference between the vehicle's orientation and the reference path orientation violates a threshold θ_{uturn}:

$$\begin{aligned}&\mathrm{makes\_u\_turn}(x_\mathrm{k}) \iff \\ &\quad \exists l \in \mathrm{occupied\_lanelets}(x_\mathrm{k}) : \theta_{\text{uturn}} < |\theta_\mathrm{k} - \mathrm{ori_l}(l, s_\mathrm{k})|.\end{aligned}$$

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